

Global E-Business and Collaboration

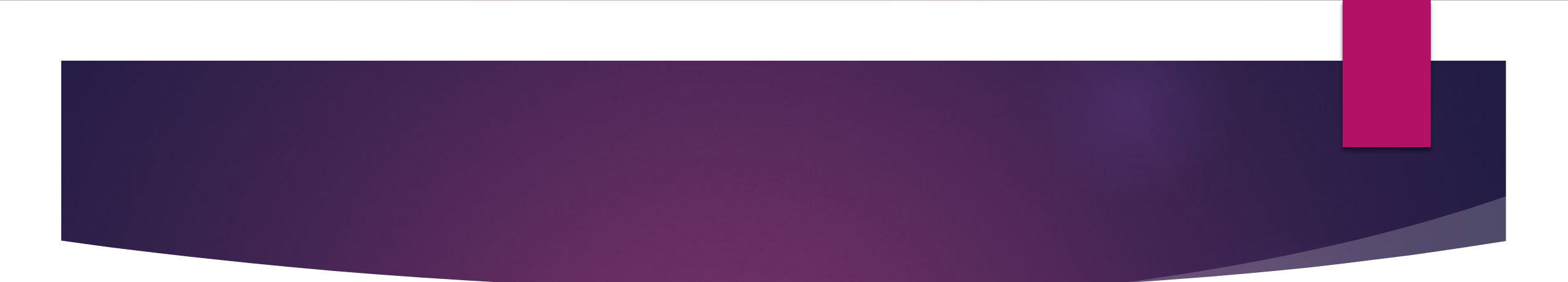
-NILAM SANGROULA

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- ▶ In order to operate, businesses must deal with many different pieces of information about suppliers, customers, employees, invoices and payments, and of course, their products and services.
- ▶ They must organize work activities that use this information to operate efficiently and enhance the overall performance of the firm. Information systems make it possible for firms to manage all their information, make better decisions, and improve the execution of their business processes.

Business Process and Information System

- ▶ Business processes refer to the manner in which work is organized, coordinated, and focused to produce a valuable product or service.
- ▶ Business processes are the collection of activities required to produce a product or service. These activities are supported by flows of material, information, and knowledge among the participants in business processes.
- ▶ Business processes also refer to the unique ways in which organizations coordinate work, information, and knowledge, and the ways in which management chooses to coordinate work.
- ▶ Every business can be seen as a collection of business processes. Business processes are designed to add value for the customer and should not include unnecessary activities.
- ▶ The outcome of a well-designed business process increased effectiveness (value for the customer) and increased efficiency (less costs for the company).

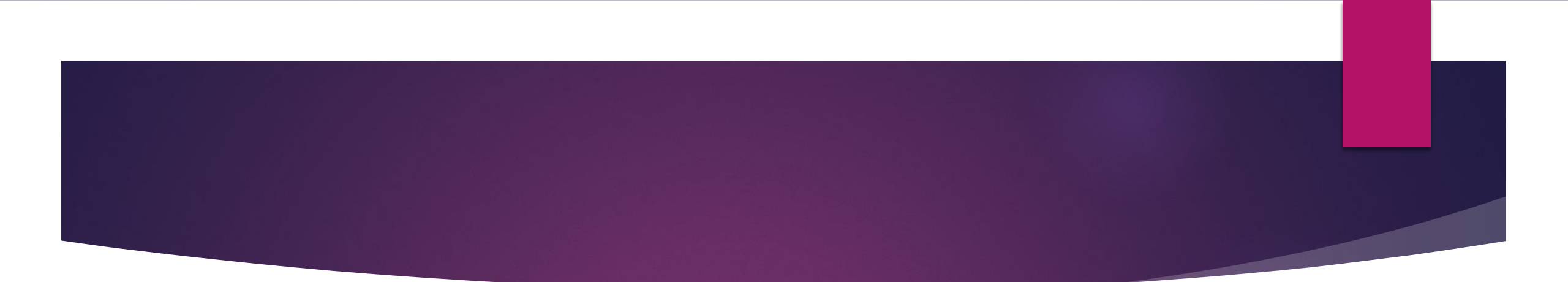
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- ▶ To a large extent, the performance of a business firm depends on how well its business processes are designed and coordinated.
 - ▶ A company's business processes can be a source of competitive strength if they enable the company to innovate or to execute better than its rivals.
 - ▶ Business processes can also be liabilities if they are based on outdated ways of working that impede organizational responsiveness and efficiency.

Types of business processes

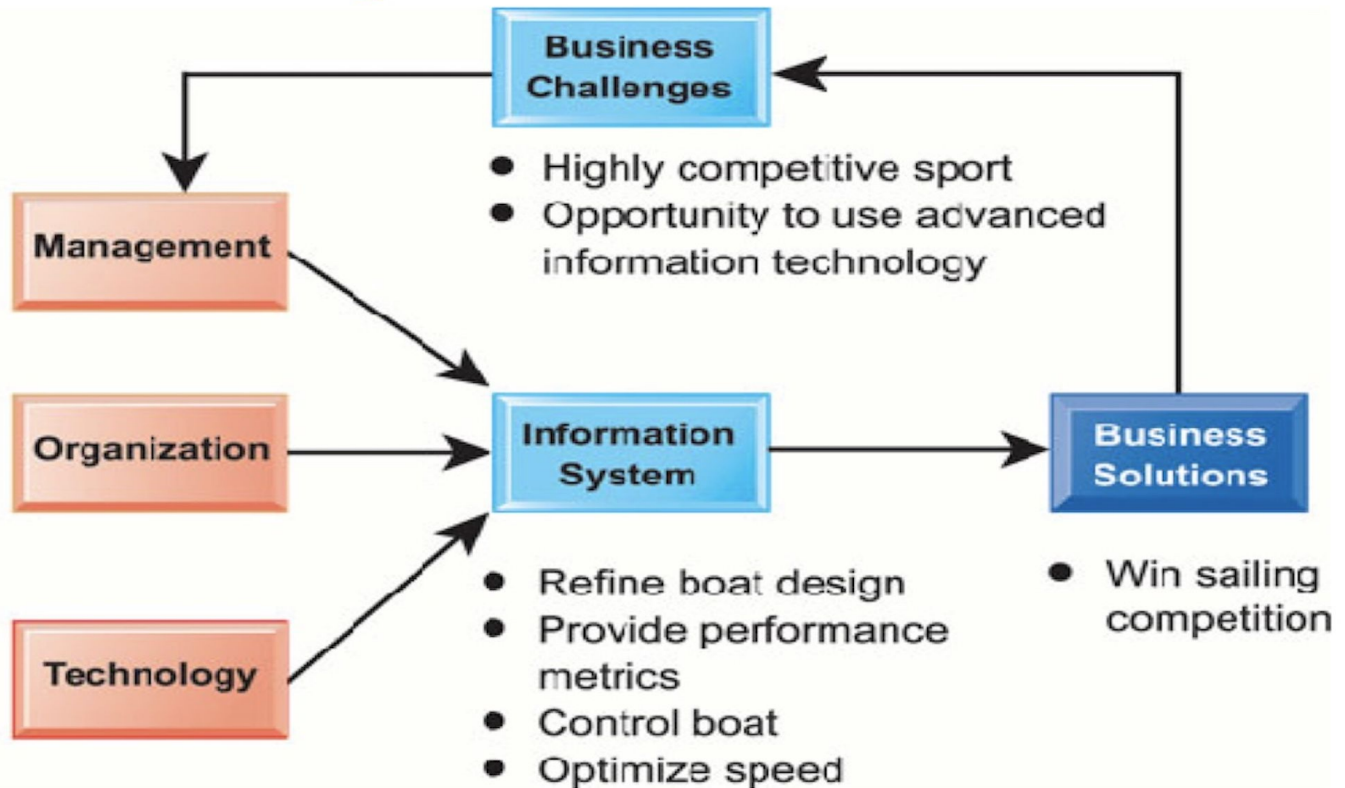
- ▶ **Management Processes:** The processes that govern the operation of a system. Typical management processes include corporate governance and strategic management
- ▶ **Operational Processes:** The processes that constitute the core business and create the primary value stream. For example, taking orders from customers, and opening an account in a bank branch.
- ▶ **Supporting Processes:** These processes support the core processes. Examples include Health & Safety, accounting, recruitment, call center, technical support etc.

Functional Areas of Business Process

Functional Area	Business Process
Manufacturing and production	Assembling the product Checking for quality Producing bills for materials
Sales and marketing	Identifying customers Making customers aware of the product Selling the product
Finance and accounting	Paying credits Creating financial statements Managing cash accounts
Human resources	Hiring employees Evaluating employees' job performance Enrolling employees in benefits plans

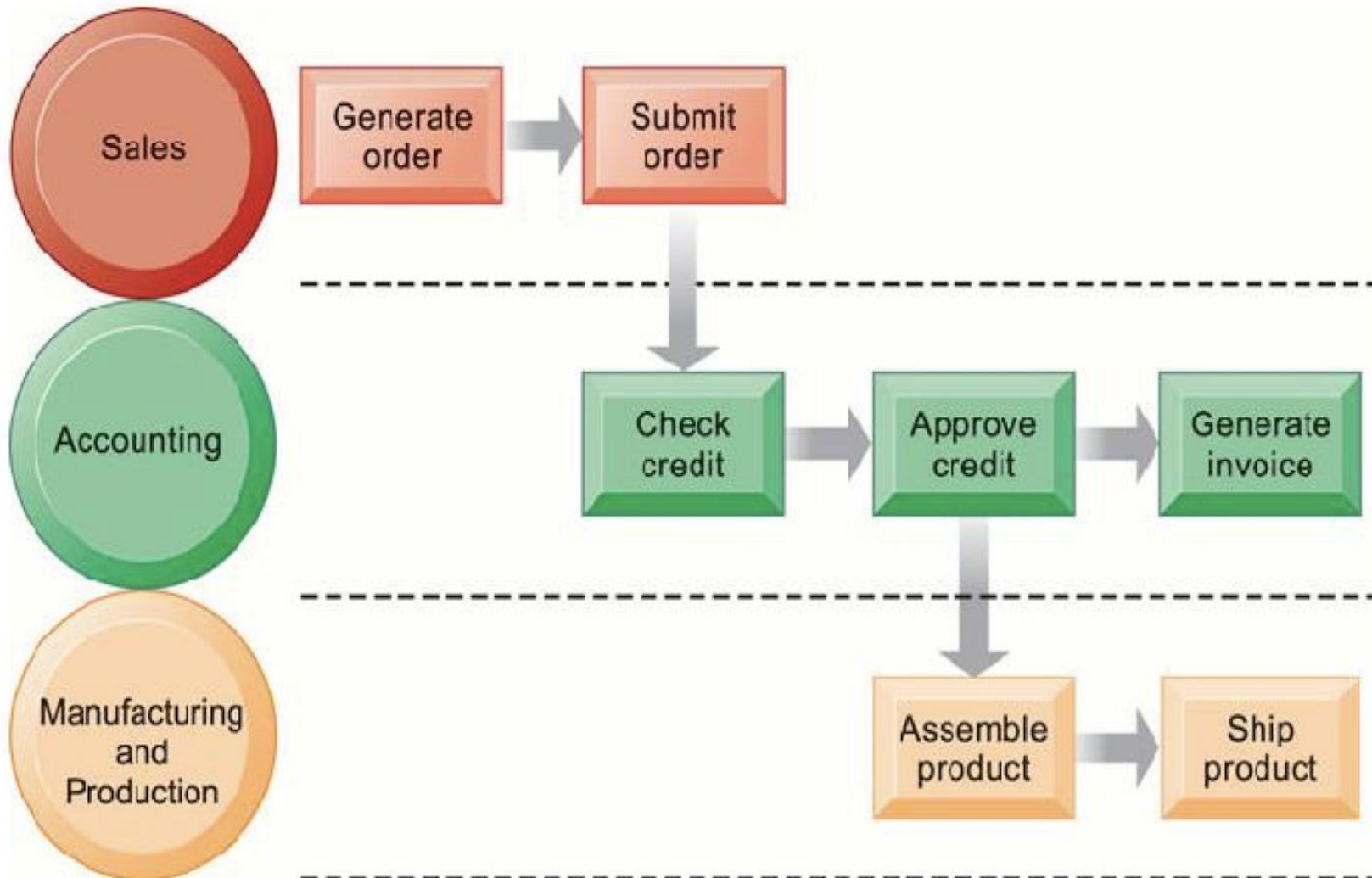
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- ▶ Oracle is one of the world's leading information technology providers, the company was a natural for using the most advanced information technology to continually improve USA's design and performance. But information technology alone would not have produced a winning boat.
 - ▶ The Oracle team had to revise many of the processes and procedures used in sailing to take advantage of the technology, including training experienced sailors to work more like pilots with high-tech instruments and sensors.
 - ▶ Oracle won the America's Cup because it had learned how to apply new technology to improve the processes of designing and sailing a competitive sailboat.

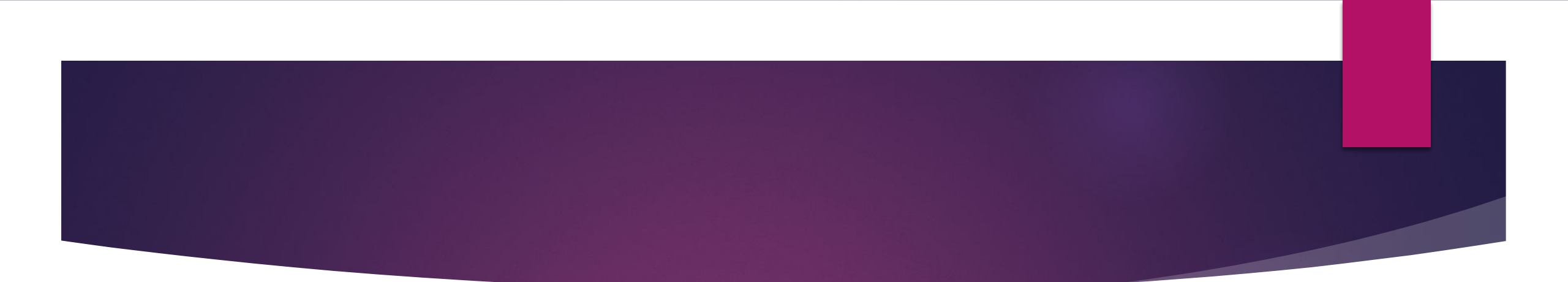
- Identify opportunity to use technology
- Develop design specifications
- Redesign sailing processes
- Train sailing team
- Deploy sensors
- Deploy Oracle data management, analytics, and graphics software
- Deploy wrist computers



How Information Technology Improves Business Processes

- ▶ Exactly, how do information systems improve business processes? Information systems automate many steps in business processes that were formerly performed manually, such as checking a client's credit, or generating an invoice and shipping order.
- ▶ But today, information technology can do much more. New technology can actually change the flow of information, making it possible for many more people to access and share information, replacing sequential steps with tasks that can be performed simultaneously, and eliminating delays in decision making.



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- ▶ New information technology frequently changes the way a business works and supports entirely new business models.
 - ▶ Downloading a Kindle e-book from Amazon, buying a computer online at Best Buy, and downloading a music track from iTunes are entirely new business processes based on new business models that would be inconceivable without today's information technology.
 - ▶ That's why it's so important to pay close attention to business processes, both in your information systems course and in your future career. By analyzing business processes, you can achieve a very clear understanding of how a business actually works.
 - ▶ Moreover, by conducting a business process analysis, you will also begin to understand how to change the business by improving its processes to make it more efficient or effective. Throughout this book, we examine business processes with a view to understanding how they might be improved by using information technology to achieve greater efficiency, innovation, and customer service.

Business Improvement with Information Technology

- ▶ Information technology enhances business processes in two main ways:
 - ▶ *Increasing efficiency of existing processes:* Automating steps or processes that were manual.
 - ▶ *Enabling entirely new processes that are capable of transforming the businesses:* This is done by changing flow of information, or by replacing sequential steps with parallel steps, or by eliminating delays in decision making.

Types of Information Systems

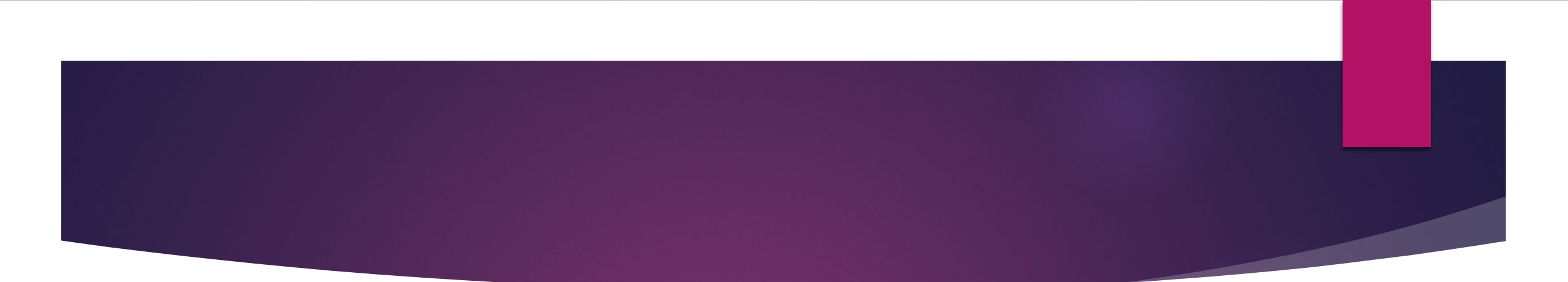
- ▶ There are different kinds of systems according to different interests, specialists, and levels in an organization. Single system cannot provide all the information needed by an organization.
- ▶ A typical business organization has systems supporting processes for each of the major business functions—systems for sales and marketing, manufacturing and production, finance and accounting, and human resources.
- ▶ The four main types of information systems that serve different management groups are:
 - ▶ Transaction processing systems
 - ▶ Management information systems
 - ▶ Decision support systems
 - ▶ Executive support systems.

Transaction Processing Systems (TPS)

- ▶ These are the computerized systems that perform and records the daily routine transactions necessary to conduct business. These systems serve the operational level of the organization.
- ▶ Some examples include sales order entry, hotel reservation systems, payroll, employee record keeping, and shipping.
- ▶ Transaction processing systems are central to a business. TPS failure for a few hours can cause a firm's demise and perhaps other firms linked to it.
- ▶ Managers need TPS to monitor the status of internal operations and the firm's relations with the external environment.
- ▶ TPS are also major producers of information for the other systems and business functions.

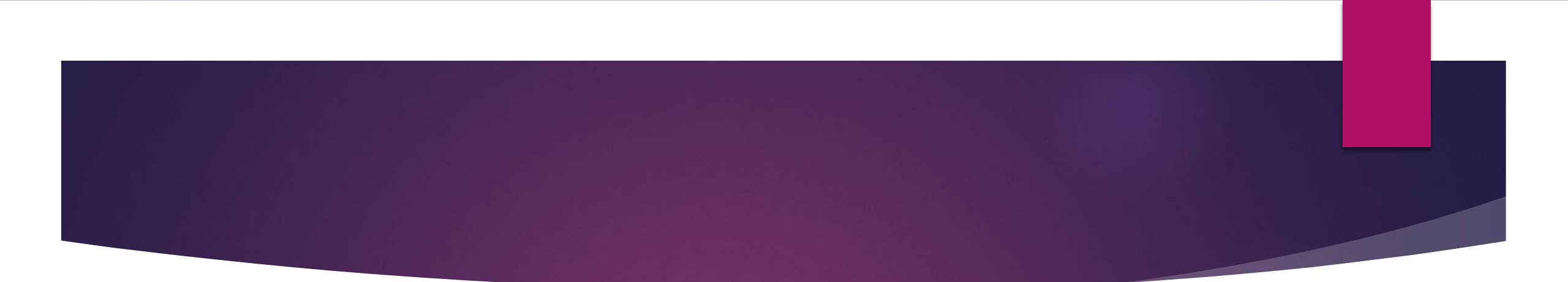
Management Information Systems (MIS)

- ▶ These are the information systems at the management level of an organization and serve management-level functions like planning, controlling, and decision-making.
- ▶ These systems provide routine summary of reports and, in some cases, with online access to the organization's current performance and historical records to managers.
- ▶ Typically, these systems use internal data provided by the transaction processing systems. Normally, these systems are used for structured decision-making.
- ▶ MIS usually serve managers interested in weekly, monthly, bimonthly results—not day-to day activities.
- ▶ MIS generally provides answers to the routine questions that have been specified in advance and have predefined procedures for answering them.

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- ▶ MIS provide middle managers with reports on the organization's current performance. This information is used to monitor and control the business and predict future performance.
 - ▶ MIS summarize and report on the company's basic operations using data supplied by transaction processing systems.
 - ▶ The basic transaction data from TPS are compressed and usually presented in reports that are produced on a regular schedule.
 - ▶ Figure below shows how a typical MIS transforms transaction-level data from order processing production, and accounting into MIS files that are used to provide managers with reports.

Decision-Support Systems (DSS):

- ▶ Decision-support systems (DSS) support more non-routine decision making.
- ▶ They focus on problems that are unique and rapidly changing, for which the procedure for arriving at a solution may not be fully predefined in advance.
- ▶ They try to answer questions such as these:
 - ▶ What would be the impact on production schedules if we were to double sales in the month of December?
 - ▶ What would happen to our return on investment if a factory schedule were delayed for six months?

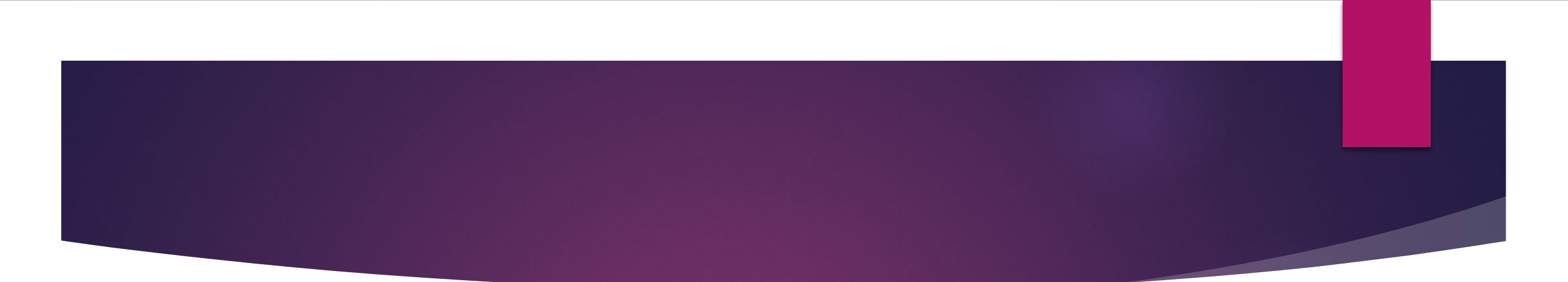
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- ▶ Although DSS use internal information from TPS and MIS, they often bring in information from external sources, such as current stock prices or product prices of competitors.
 - ▶ These systems use a variety of models to analyze the data and are designed so that users can work with them directly.
 - ▶ DSS have more analytical power than other systems. DSS helps managers to make decision that are unique, fast changing and not easily specified in advance. DSS that supports and facilitates the decision-making process by a group is called group decision support system (GDSS).

Quantitative models used by DSS

- ▶ **Sensitivity analysis** is the study of the impact that changes in one (or more) parts of the model have on other parts of the model. Users change the value of one variable repeatedly and observe the resulting changes in other variables.
- ▶ **What-if analysis** checks the impact of a change in an assumption on the proposed solution. For example, —What will happen to the supply chain if a Bhatbhateni reduces holding inventory from 30 percent to 10 percent? Users repeat this analysis until they understand all the effects of various situations.
- ▶ **Goal-seeking analysis** finds the inputs necessary to achieve a goal such as a desired level of output. Instead of observing how changes in a variable affect other variables as in what-if analysis, goal-seeking analysis sets a target value (a goal) for a variable and then repeatedly changes other variables until the target value is achieved. For example, —How many customers are required to purchase a new product to increase gross profits to \$5 million?

Executive Support Systems (ESS)

- ▶ These systems serve the strategic level of the organization. These systems are designed to address unstructured decision making through advanced graphics and communication.
- ▶ These systems incorporate data about external events such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS.
- ▶ These systems are not designed to solve a specific problem but they provide a generalized computing and telecommunication capacity that can be applied to a changing array of problems.
- ▶ ESS helps senior executives to monitor firm performance, spot problems, identify opportunities, and forecast trends.

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- ▶ These systems can filter out extraneous details for high level overviews, or they can drill down to provide senior managers to detailed transaction data if required.
 - ▶ ESS helps senior managers to analyze, compare, and highlight trends so that they may easily monitor organizational performance or identify strategic problem and opportunities.
 - ▶ There is less use of analytical tools (as compared to DSS) in ESS.
 - ▶ It is not necessary for users to be an expert in computer-based information system to be able to use them. Therefore, executive system must be easy to use and the information must be easily manipulated.

Systems for Linking the Enterprise

- ▶ Reviewing all the different types of systems we have just described, anyone might wonder how a business can manage all the information in these different systems.
- ▶ People might wonder how costly it is to maintain so many different systems and how all these different systems can share information and how managers and employees are able to coordinate their work. In fact, these are all important questions for businesses today.

Enterprise Systems

- ▶ Firms use enterprise systems, also known as enterprise resource planning (ERP) systems, to integrate business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources into a single software system. Information that was previously fragmented in many different systems is stored in a single comprehensive data repository where it can be used by many different parts of the business.
- ▶ For example, when a customer places an order, the order data flow automatically to other parts of the company that are affected by them. The order transaction triggers the warehouse to pick the ordered products and schedule shipment. The warehouse informs the factory to replenish whatever has been depleted. The accounting department is notified to send the customer an invoice. Customer service representatives track the progress of the order through every step to inform customers about the status of their orders. Managers are able to use firm-wide information to make more precise and timely decisions about daily operations and longer-term planning.

Supply Chain Management Systems

- ▶ Firms use supply chain management (SCM) systems to help manage relationships with their suppliers. These systems help suppliers, purchasing firms, distributors, and logistics companies share information about orders, production, inventory levels, and delivery of products and services so that they can source, produce, and deliver goods and services efficiently. The ultimate objective is to get the right amount of their products from their source to their point of consumption in the least amount of time and at the lowest cost. These systems increase firm profitability by lowering the costs of moving and making products and by enabling managers to make better decisions about how to organize and schedule sourcing, production, and distribution.
- ▶ Supply chain management systems are one type of inter organizational system because they automate the flow of information across organizational boundaries.

Customer Relationship Management Systems

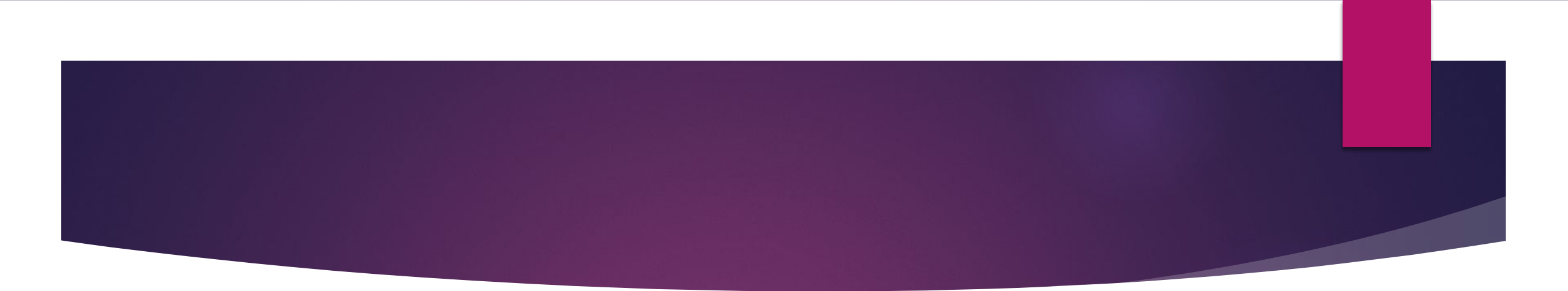
- ▶ Firms use customer relationship management (CRM) systems to help manage their relationships with their customers. CRM systems provide information to coordinate all of the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention.
- ▶ This information helps firms identify, attract, and retain the most profitable customers; provide better service to existing customers; and increase sales.

Knowledge Management Systems

- ▶ Knowledge management (KM) refers to the systematic process of capturing, organizing, sharing, and analyzing the knowledge and information within an organization. It aims to enhance organizational efficiency and innovation by ensuring that the right information is accessible to the right people at the right time.
- ▶ Some firms perform better than others because they have better knowledge about how to create, produce, and deliver products and services. This firm knowledge is difficult to imitate, unique, and can be leveraged into long-term strategic benefits.
- ▶ Knowledge management systems (KMS) enable organizations to better manage processes for capturing and applying knowledge and expertise. These systems collect all relevant knowledge and experience in the firm, and make it available wherever and whenever it is needed to improve business processes and management decisions. They also link the firm to external sources of knowledge.

Systems for Collaboration and Teamwork

- ▶ Collaboration is working with others to achieve shared and explicit goals. Collaboration focuses on task or mission accomplishment and usually takes place in a business, or other organization, and between businesses.
- ▶ You collaborate with a colleague in Tokyo having expertise on a topic about which you know nothing. You collaborate with many colleagues in publishing a company blog. If you're in a law firm, you collaborate with accountants in an accounting firm in servicing the needs of a client with tax problems.
- ▶ Collaboration can be short-lived, lasting a few minutes, or longer term, depending on the nature of the task and the relationship among participants. Collaboration can be one-to-one or many-to-many.

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- ▶ Collaboration and teamwork are more important today than ever for a variety of reasons.
 - ▶ Changing nature of work
 - ▶ Growth of professional work
 - ▶ Changing organization of the firm
 - ▶ Changing scope of the firm
 - ▶ Emphasis on innovation
 - ▶ Changing culture of work and business

Business Benefits of Collaboration and Teamwork

BENEFIT	RATIONALE
Productivity	People working together can complete a complex task faster than the same number of people working in isolation from one another. There will be fewer errors.
Quality	People working collaboratively can communicate errors, and correct actions faster, when they work together than if they work in isolation. Can lead to a reduction in buffers and time delay among production units.
Innovation	People working collaboratively in groups can come up with more innovative ideas for products, services, and administration than the same number working in isolation from one another.
Customer Service	People working together in teams can solve customer complaints and issues faster and more effectively than if they were working in isolation from one another.
Financial performance	As a result of all of the above, collaborative firms have superior sales growth and financial performance.

Requirements for Collaboration

Collaboration Capability

- Open culture
- Decentralized structure
- Breadth of collaboration

Collaboration Technology

- Use of collaboration technology for implementation and operations
- Use of collaborative technology for strategic planning

Collaboration Quality

Firm Performance

